

VIETNAM NATIONAL UNIVERSITY-HO CHI MINH CITY

HO CHI MINH UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

KNOWLEDGE ENGINEERING DEPARTMENT

Homework 3

Course: Introduction to Artificial Intelligence

Student:

23127165 - Nguyen Hai Dang

Instructors:

Mr Bui Tien Len

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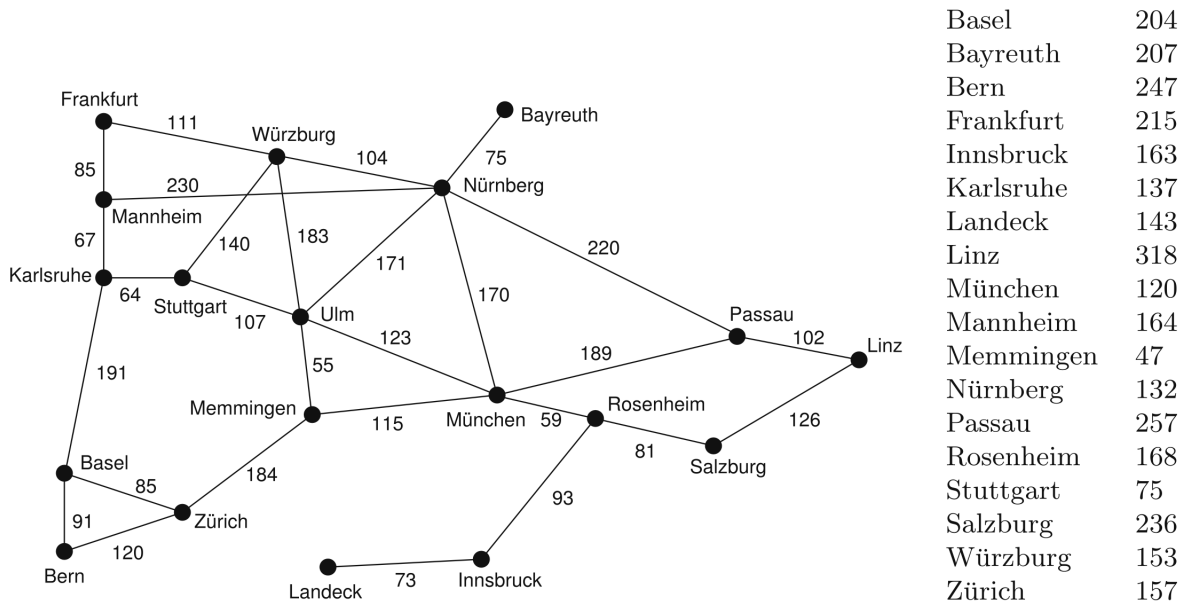
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1 Exercise

Given German map, for each of the following **graph search** strategies, find the path from Linz to Ulm and list the order in which cities are expanded, as well as the path returned by graph search. (Note that use alphabet order for successors)

flying distances from all cities to Ulm



1. Greedy best-first search
2. A*

2 Greedy best-first search

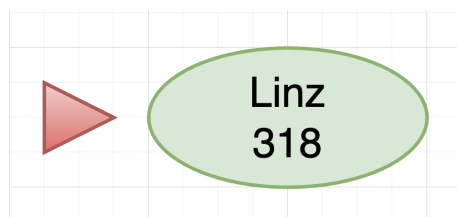
2.1 Algorithm for Graph Search

- Greedy Best-First Search works by evaluating the heuristic cost $h(n)$ of each node and expanding the node with the smallest $h(n)$. This process is repeated until the goal is reached.
- To avoid revisiting nodes, the algorithm uses an explored set to track expanded nodes.
- A priority queue is used to order nodes by their heuristic value.
- The heuristic function $h(n)$ should be admissible and consistent for optimal results. In this problem, $h(n)$ is defined as the straight-line distance from each city to Ulm.

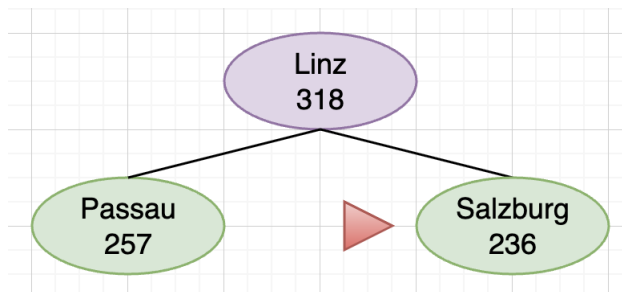
2.2 Implementation for Graph Search

- (*)**Green nodes:** Represent frontier nodes (discovered but not yet expanded).
- (*)**Violet nodes:** Indicate expanded nodes (fully explored).

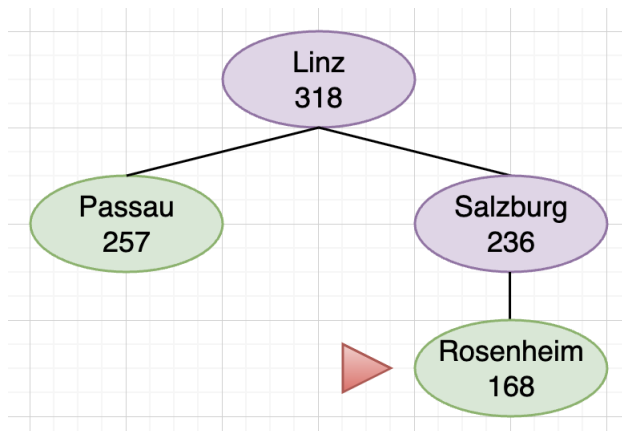
- Step 1: The initial state



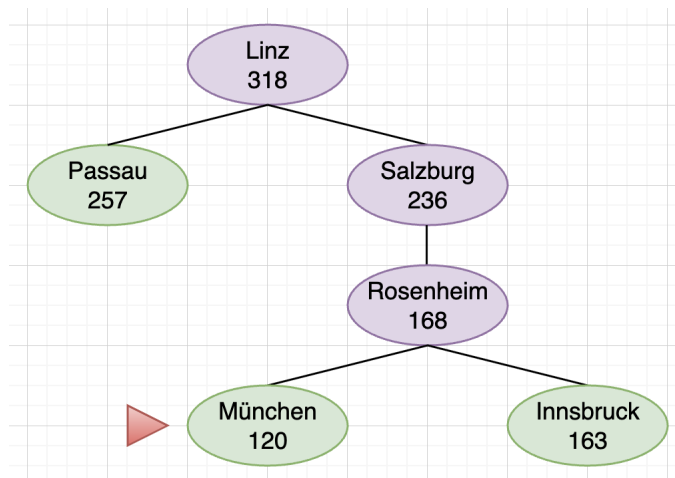
- Step 2: After expanding Linz



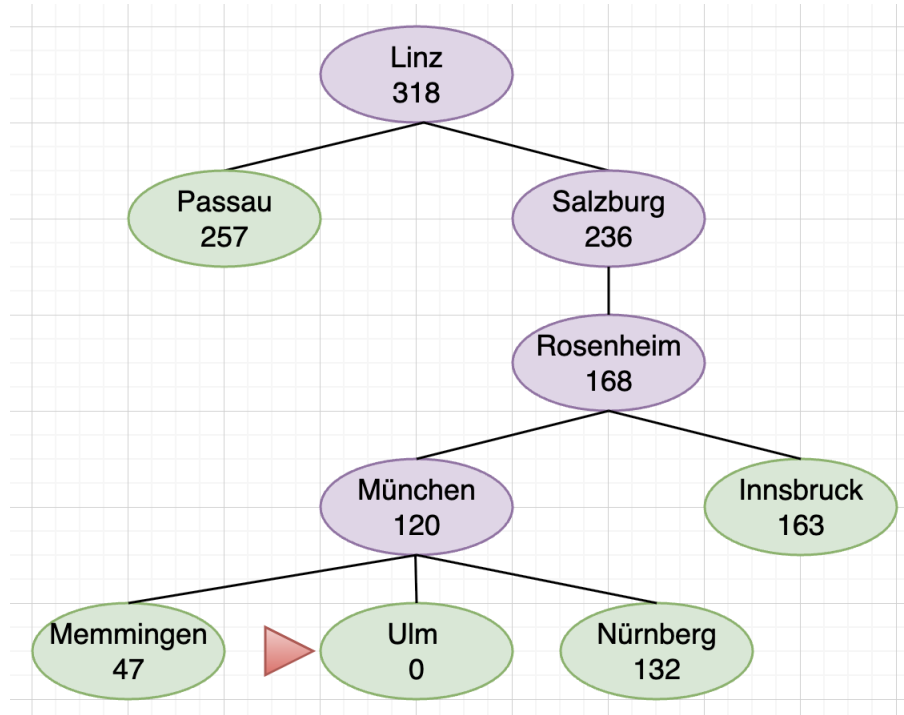
- Step 3: After expanding Salzburg



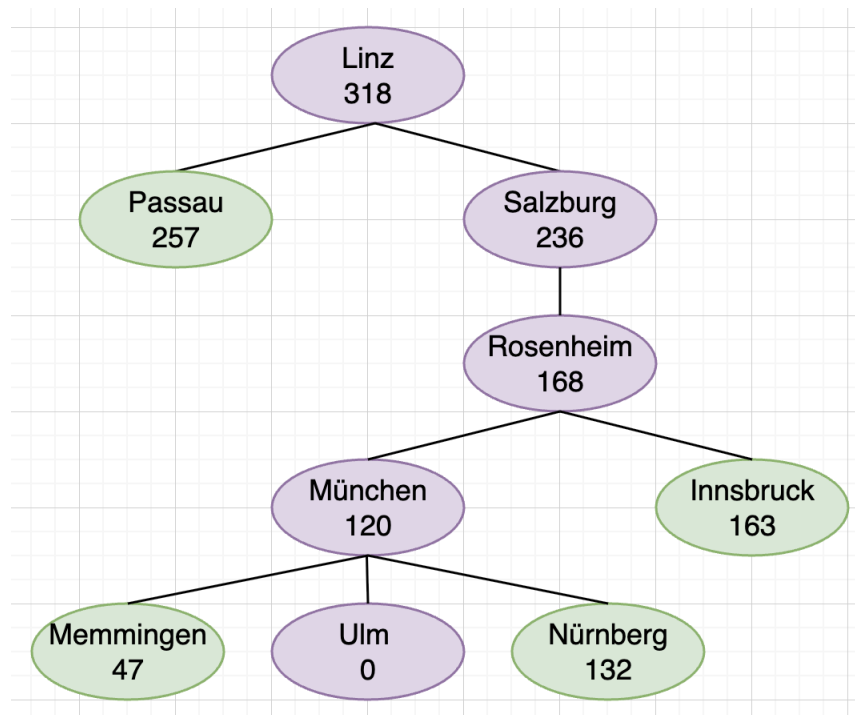
- Step 4: After expanding Rosenheim



- Step 5: After expanding München



- Step 6: Reach the goal (Ulm)



⇒ The path returned by graph search: Linz → Salzburg → Rosenheim → München → Ulm
⇒ Path cost: $126 + 81 + 59 + 123 = 389$

3 A*

3.1 Algorithm for Graph Search

- A* search (pronounced A-star search) is a widely used form of best-first search that guarantees finding the optimal solution if certain conditions are met.

- The algorithm uses an evaluation function:

$$f(n) = g(n) + h(n)$$

- $g(n)$: The actual cost to reach node n from the start node.
- $h(n)$: The estimated cost of the cheapest path from node n to the goal (heuristic function).
- $f(n)$: The estimated total cost of the cheapest solution passing through n .

- A* prioritizes nodes with the smallest $f(n)$, effectively balancing exploration (via $g(n)$) and exploitation (via $h(n)$).

- To avoid revisiting nodes and infinite loops in graph search, A* maintains an explored set and updates nodes in the frontier if a cheaper path is found.

- For correctness, the heuristic function $h(n)$ must satisfy:

- **Admissibility:** $h(n) \leq h^*(n)$, where $h^*(n)$ is the true cost from n to the goal.
- **Consistency:** $h(n) \leq c(n, n') + h(n')$, where $c(n, n')$ is the cost between n and n' .

- In this problem:

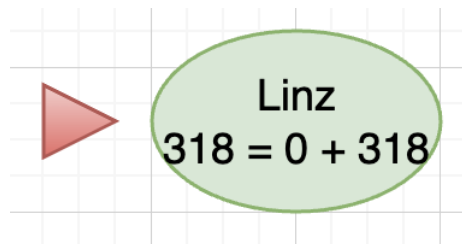
- $g(n)$: The total path cost from Linz to n .
- $h(n)$: The straight-line flying distance from n to Ulm.
- $f(n)$: Combines both to estimate the most promising path.

3.2 Implementation for Graph Search

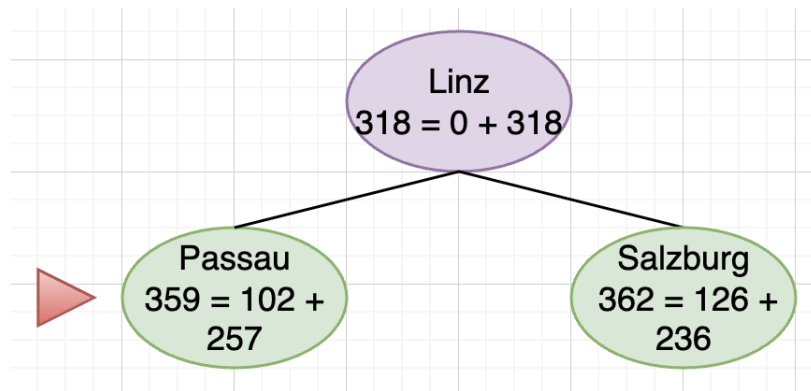
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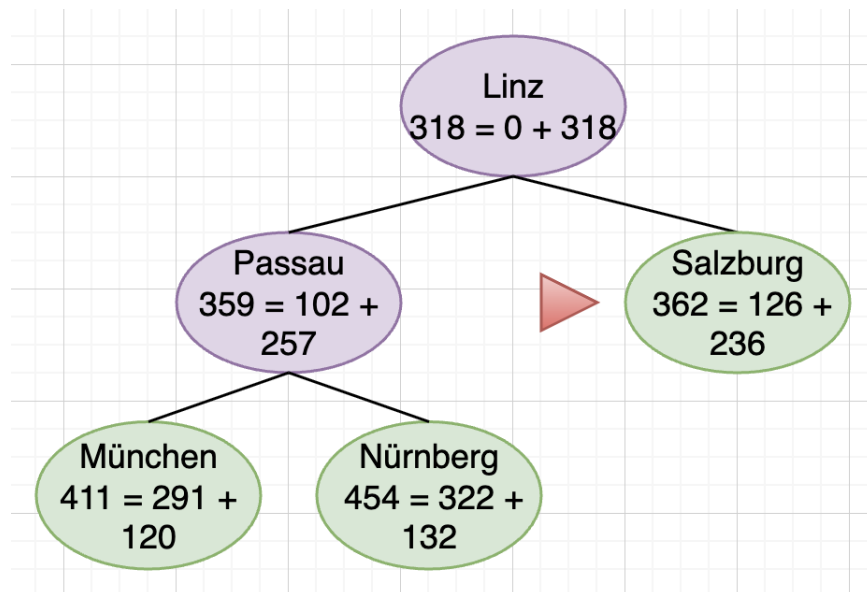
- Step 1: The initial state



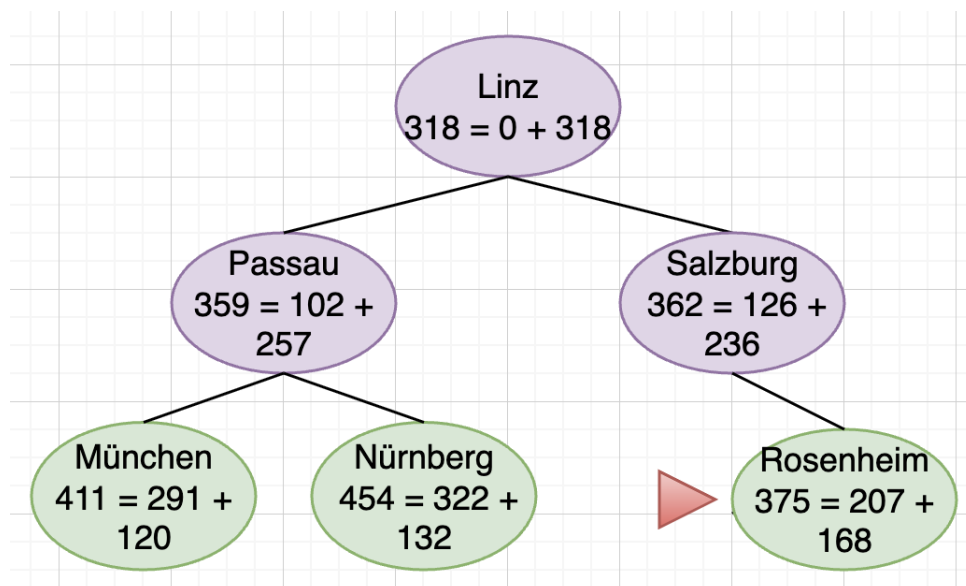
- Step 2: After expanding Linz



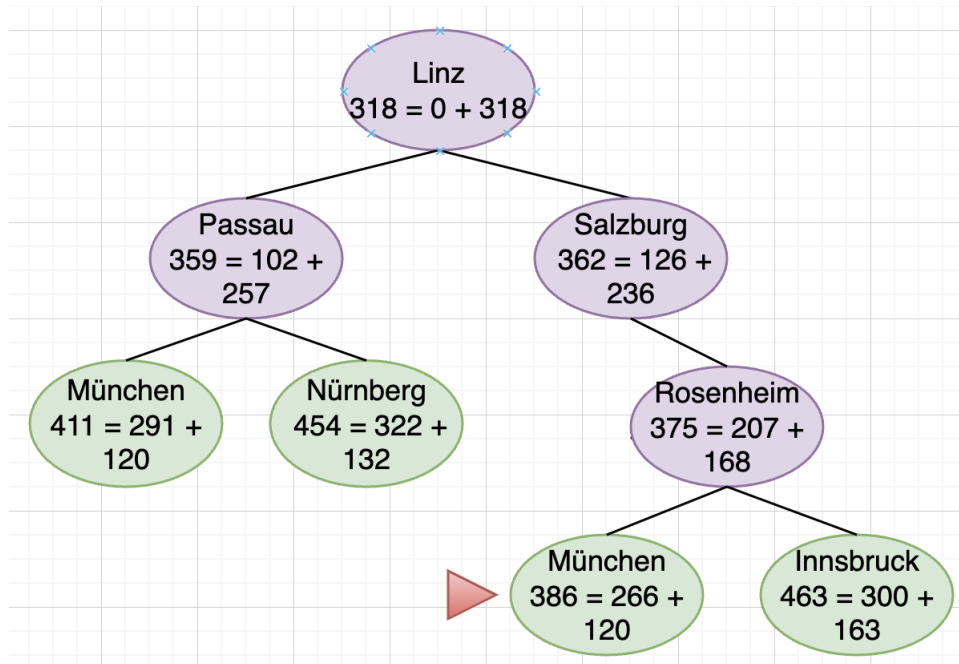
- Step 3: After expanding Passau



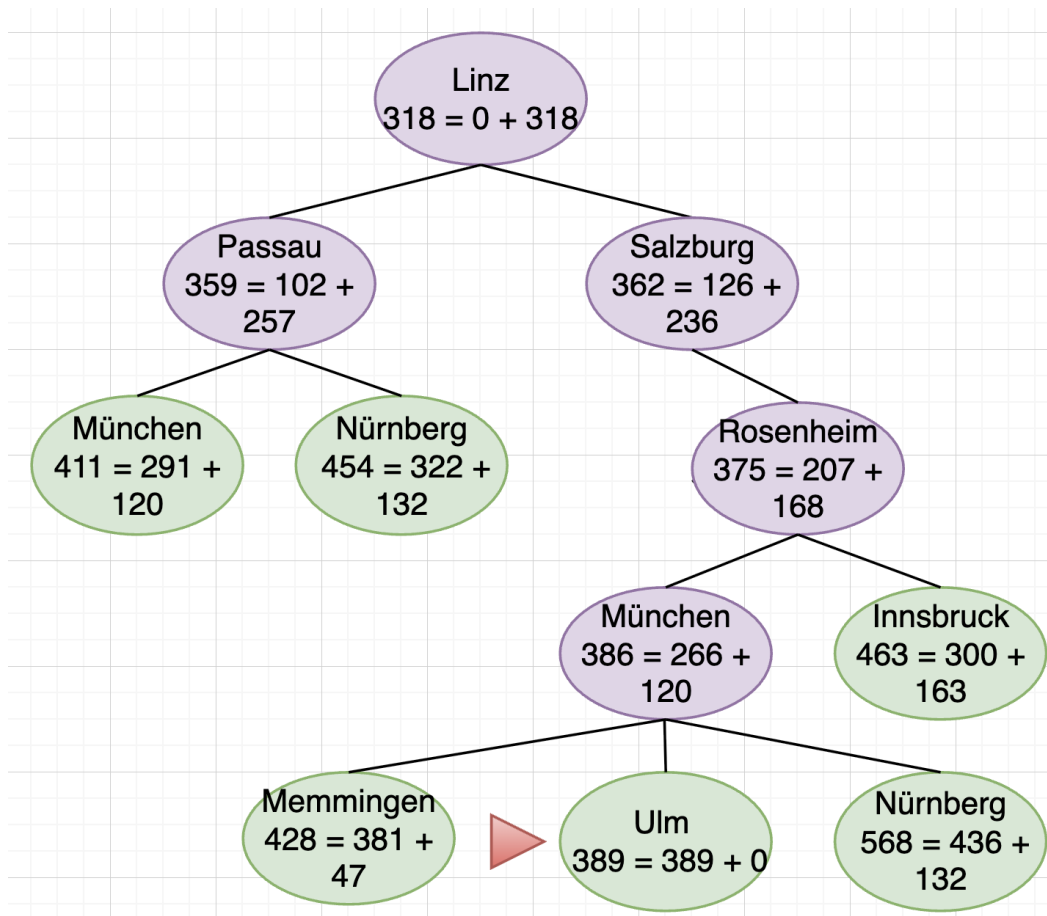
- Step 4: After expanding Salzburg



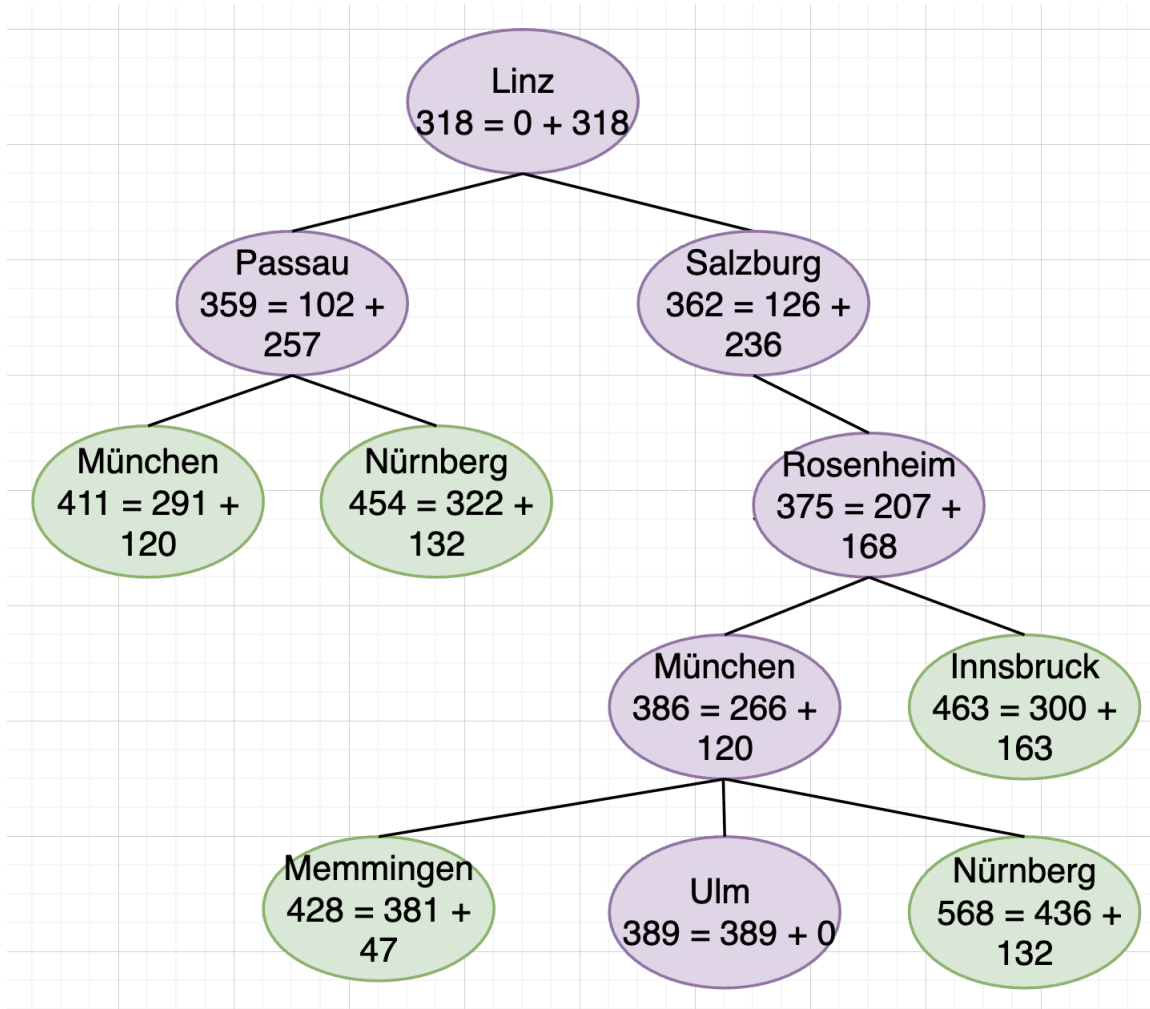
- Step 5: After expanding Rosenheim



- Step 6: After expanding München



- Step 7: Reach the goal (Ulm)



⇒ The path returned by graph search: Linz → Salzburg → Rosenheim → München → Ulm

⇒ Path cost: $126 + 81 + 59 + 123 = 389$